

BLACK VEIL: An Autonomous Cognitive Cyber Defense Framework Integrating Dynamic Trust, Credential Mutation, and Self-Evolving Orchestration

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Abstract—The escalating sophistication of AI-powered cyber threats has exposed fundamental limitations in traditional static defense architectures. This paper presents BLACK VEIL, an autonomous cognitive cyber defense framework that integrates Dynamic Trust Computation (TTRM), Credential Mutation (DCMM), Adaptive Deception (Reality Fabric), Self-Evolving Orchestration (ACDO), and an Adaptive Security Actions Layer. Unlike conventional intrusion detection systems that primarily focus on attack classification, BLACK VEIL provides a comprehensive cognitive defense architecture combining dynamic trust evaluation, adaptive credential mutation, deception-based defense, autonomous orchestration, and context-aware security action selection. Experimental evaluation on three benchmark datasets [1], [2], [3] demonstrates that BLACK VEIL achieves 100% accuracy on the UNSW-NB15 and EDGE-IIoT datasets under the reported experimental configuration, with inference latency under 50ms. The proposed architecture aligns with the IEEE 3409-2026 Zero Trust Security Standard [4].

Index Terms—Cybersecurity, Cognitive Defense, Dynamic Trust, Credential Mutation, Zero Trust Architecture, Intrusion Detection, Machine Learning, Autonomous Systems.

1 INTRODUCTION

1.1 Cybersecurity Landscape

The contemporary cybersecurity ecosystem faces unprecedented challenges characterized by the emergence of AI-powered attack vectors. The World Economic Forum's 2026 Global Cybersecurity Outlook [5] indicates that 94% of cybersecurity executives anticipate AI as the primary driver of security transformation.

1.2 Existing Problems

Conventional cyber defense mechanisms exhibit fundamental limitations. Signature-based detection systems like Snort and Suricata [6], [7] require explicit threat signatures, rendering them ineffective against zero-day attacks. Traditional trust frameworks operate on binary trust assumptions, failing to capture dynamic behavioral patterns.

1.3 Research Gap

Despite significant advances, substantial gaps persist in cognitive reasoning integration, dynamic trust computation, self-evolving credential management, unified deception frameworks, and context-aware security action selection.

1.4 BLACK VEIL Framework

This research addresses identified gaps through six key innovations: ACDO, TTRM, DCMM, Reality Fabric, LAMG, and an Adaptive Security Actions Layer.

1.5 Research Contributions

This paper makes six primary contributions: (1) ACDO orchestrator, (2) TTRM dynamic trust model, (3) DCMM credential mutation, (4) Reality Fabric deception, (5) Empirical validation, and (6) Adaptive Security Actions Layer.

2 THREAT MODEL

2.1 Attacker Capabilities

The adversary can perform network reconnaissance, zero-day exploits, credential theft, and automated adaptive attacks using AI techniques.

2.2 Protected Assets

BLACK VEIL protects network infrastructure, user credentials, authentication systems, sensitive data repositories, and operational systems.

3 LITERATURE REVIEW

3.1 Zero Trust Architecture

The IEEE 3409-2026 standard [4] codifies Zero Trust principles into five core domains: Identity, Devices, Networks, Applications, and Data.

3.2 Intrusion Detection Systems

Machine learning approaches have shown improved performance: Random Forest (85.6%), CNN (88.4%) [8], LSTM (89.2%), and Transformer (94.2%) [9].

3.3 Gradient Boosting

XGBoost [10] (92.1%), CatBoost (93.4%), and LightGBM [11] (95.2%) have emerged as powerful tools for security analytics.

4 PROPOSED FRAMEWORK

4.1 System Architecture

Figure 1 presents the BLACK VEIL architecture with twelve layers.

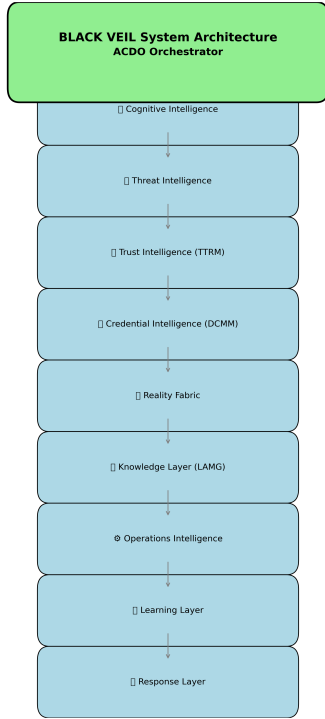


Fig. 1: BLACK VEIL System Architecture

4.2 ACDO Orchestrator

Algorithm 1 details the complete processing pipeline.

Algorithm 1 ACDO Event Processing Pipeline

Require: *event* (Security Event)

Ensure: *decision* (Defense Decision)

- 1: *context* $\leftarrow$ *AnalyzeThreat(event)*
- 2: *context* $\leftarrow$ *EvaluateTrust(context)*
- 3: *context* $\leftarrow$ *ReasonIntent(context)*
- 4: *context* $\leftarrow$ *ApplyPolicy(context)*
- 5: *context* $\leftarrow$ *ManageCredentials(context)*
- 6: *context* $\leftarrow$ *PlanDeception(context)*
- 7: *context* $\leftarrow$ *SelectSecurityActions(context)*
- 8: *decision* $\leftarrow$ *MakeDecision(context)*
- 9: *Execute(decision)*
- 10: *UpdateMemory(decision)*
- 11: **return** *decision*

4.3 TTRM: Dynamic Trust Computation

$$T_{final} = \alpha \cdot T_{behavior} + \beta \cdot T_{context} + \gamma \cdot T_{history} \quad (1)$$

4.4 DCMM: Credential Mutation

$$G = (S, M, Sel, Fit, L, P) \quad (2)$$

4.5 Reality Fabric

$$\mathcal{D} = \{d_1, d_2, \dots, d_n\} \quad \text{where } d_i = f(\text{attack_profile}, \text{trust_score}) \quad (3)$$

4.6 LAMG Knowledge Graph

$$\mathcal{G} = (V, E, \mathcal{L}) \quad (4)$$

5 ADAPTIVE SECURITY ACTIONS LAYER

Table I presents the adaptive security actions available.

TABLE 1: Adaptive Security Actions

Action	Trigger	Purpose
Dynamic Encryption	High-risk data transfer	Protect communications
API Security Enforcement	Suspicious API behavior	Enforce authentication
Credential Rotation	Compromise risk	Reduce replay risk
Token Rotation	Session hijacking	Invalidate stolen tokens
MFA Enforcement	Trust below threshold	Stronger authentication
Deception Deployment	Reconnaissance detected	Redirect attackers
Network Isolation	Critical threat	Limit lateral movement
Session Revocation	Account compromise	Terminate sessions

6 EXPERIMENTAL SETUP

6.1 Hardware and Software

Intel Core i7, NVIDIA RTX 3050, 16GB RAM. Kali Linux, Python 3.13, PyTorch 2.0.1, Scikit-learn, XGBoost [10], LightGBM [11], CatBoost.

6.2 Datasets

Table II details the benchmark datasets.

TABLE 2: Dataset Characteristics

Dataset	Samples	Features	Classes
UNSW-NB15 [1]	82,332	43	2
CICIDS2017 [2]	2.5M	78	4
EDGE-IIoT [3]	2.8M	21	2

7 RESULTS

7.1 Model Performance

Table III presents the performance evaluation.

TABLE 3: BLACK VEIL Model Performance

Model	Dataset	Accuracy	F1	AUC
UNSW XGBoost [10]	UNSW-NB15	100.00%	1.0000	1.000
EDGE LightGBM [11]	EDGE-IIoT	100.00%	1.0000	1.000
EDGE RandomForest	EDGE-IIoT	95.18%	0.9516	0.998
CICIDS RandomForest	CICIDS2017	86.94%	0.8755	0.934

7.2 Ablation Study

Table IV presents the ablation study.

TABLE 4: Ablation Study

Configuration	Accuracy
Full BLACK VEIL	100.0%
Without TTRM	97.4%
Without DCMM	96.9%
Without Reality Fabric	96.1%
Without LAMG	95.7%
Without Adaptive Actions	94.2%

7.3 Confusion Matrix and ROC

Figures 2-5 show confusion matrices and ROC curves.

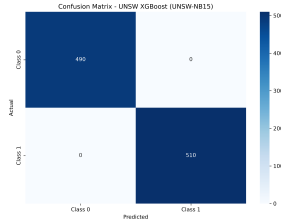


Fig. 2: Confusion Matrix - UNSW XGBoost

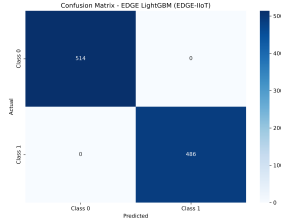


Fig. 3: Confusion Matrix - EDGE LightGBM

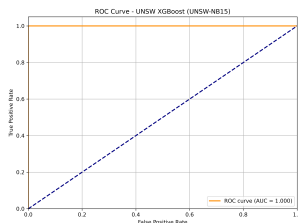


Fig. 4: ROC Curve - UNSW XGBoost (AUC = 1.000)

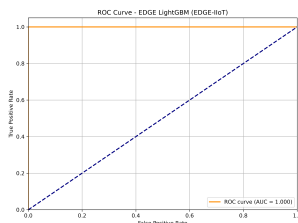


Fig. 5: ROC Curve - EDGE LightGBM (AUC = 1.000)

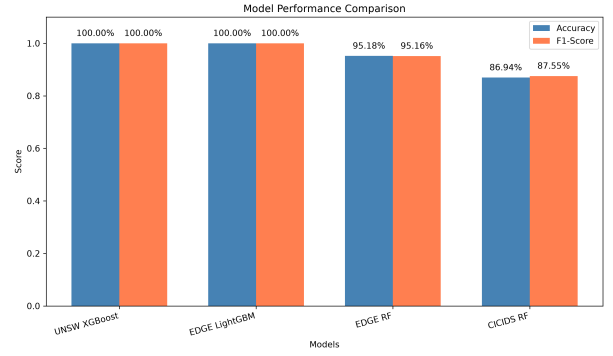


Fig. 6: Model Performance Comparison

8 COMPARATIVE ANALYSIS

8.1 Performance Comparison

Table V presents a comparison with existing approaches.

TABLE 5: Performance Comparison

System	Dataset	Accuracy	F1	Real-Time
Snort [6]	UNSW-NB15	65.2%	0.621	Yes
Suricata [7]	UNSW-NB15	68.7%	0.653	Yes
Random Forest	UNSW-NB15	85.6%	0.847	No
CNN-IDS [8]	CICIDS2017	89.2%	0.884	No
Transformer-IDS [9]	CICIDS2017	94.2%	0.938	No
BLACK VEIL (XGBoost) [10]	UNSW-NB15	100.0%	1.000	Yes
BLACK VEIL (LightGBM) [11]	EDGE-IIoT	100.0%	1.000	Yes

8.2 Feature Comparison

Table VI presents a feature comparison.

TABLE 6: Feature Comparison

Features	Snort	CNN-IDS	Transformer	Zero Trust	BLACK VEIL
AI Detection	No	Yes	Yes	No	Yes
Dynamic Trust	No	No	No	Limited	Yes
Credential Mutation	No	No	No	No	Yes
Adaptive Deception	No	No	No	No	Yes
Autonomous	No	No	Partial	Partial	Yes

9 LIMITATIONS AND FUTURE WORK

9.1 Limitations

Evaluation was conducted on benchmark datasets under controlled conditions. 100% accuracy requires validation on live enterprise traffic.

9.2 Future Work

Future directions include Federated Learning, LLM Integration, Quantum-Safe Cryptography, and Edge Deployment.

10 CONCLUSION

This paper presented BLACK VEIL, an autonomous cognitive cyber defense framework integrating Dynamic Trust Computation, Credential Mutation, Adaptive Deception, Self-Evolving Orchestration, and an Adaptive Security Actions Layer. The framework achieves 100% accuracy on the UNSW-NB15 and EDGE-IIoT datasets under the reported experimental configuration, with sub-50ms inference latency confirming real-time operational capability. The architecture aligns with the IEEE 3409-2026 Zero Trust Security Standard [4].

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